1. Clinical Question & Core Hypothesis

Clinical Question

TGF-β–driven fibrotic microenvironments in MIBC (which should create immune exclusion) do not uniformly predict lack of benefit from immunotherapy — some fibrotic tumors nevertheless respond. Why and under what molecular conditions is immune exclusion by fibrosis overcome? 2) Tier 1 paths used: CORE PATH 1 (TGFB1/TGF-β) provides the mechanistic axis for fibrosis/immune exclusion and links directly to a histologic stromal phenotype (TSR); CORE PATH 2 (MSH6) supplies a genomic-immune counterweight (MMR deficiency → higher neoantigen load) that can recruit immune effectors despite exclusionary stroma; CORE PATH 3 (urine DNA methylation) was acknowledged as an important noninvasive readout but deliberately not placed in the core variables to keep the mechanistic hypothesis sharp. 3) Tier 2 evidence supporting the claim: metabolic markers (HDL-C, ALP, TG, SHBG, IGF1) can modulate systemic inflammation and T cell function and therefore may influence whether neoantigen-rich tumors generate effective immunity; AI-based biomarker approaches can integrate TSR, mutational/MMR signals, and methylation readouts to stratify response; SDC2/TFPI2 methylation shows the feasibility of epigenetic, noninvasive tumor detection (analogous to urine methylation panels). 4) Why therapy_response as endpoint: the mechanistic tension (fibrosis-driven immune exclusion vs. neoantigen-driven immune recruitment) is most directly relevant to predicting sensitivity to immune checkpoint blockade — a clinically actionable therapy in MIBC. 5) What I excluded and why: I excluded urine DNA methylation from the two-anchor core to obey the 2-anchor-gene limit and keep a single, testable mechanistic question; Tier2 metabolic and AI factors were relegated to secondary_modifiers to be evaluated as modulators rather than core drivers; Tier3 exploratory items (pZVR, CSF AD biomarkers, PTGDS) were omitted from the hypothesis because they lack direct mechanistic grounding in MIBC and would dilute focus?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), a high stromal-to-tumor ratio (TSR) denotes a TGFB1-driven fibrotic microenvironment that creates immune exclusion and reduces anti-tumor immunity, contributing to resistance to immune checkpoint blockade (ICB). However, concurrent loss or deficiency of MSH6 (reflecting mismatch repair deficiency with elevated neoantigen burden) can partially overcome this exclusion by enhancing neoantigen-driven T cell recruitment and reactivity, generating targetable yet dysfunctional T cells. Therefore, among TSR-high tumors, those with MSH6 deficiency will exhibit higher objective response rates, longer durations of response, and improved progression-free survival when treated with ICB compared to TSR-high tumors with intact MSH6. In TSR-low tumors, the predictive power of MSH6 status is reduced, and responses to ICB are more favorable overall, reflecting lower fibrosis-associated exclusion and more accessible antitumor immunity. Urine DNA methylation markers and other Tier-2 modifiers (e.g., HDL-C, ALP, TG, SHBG, IGF1, AI-based biomarker signatures) should be evaluated as secondary modifiers to refine response predictions but are not required for the core mechanism. The anchor genes remain TGFB1 (fibrosis/TSR mediator) and MSH6 (MMR/neoantigen burden). The primary endpoints will be therapy_response, with secondary endpoints including duration of response and progression-free survival. Feasible testing approaches include retrospective analyses integrating digital pathology-based TSR assessment, MSH6 status via sequencing/IHC, and ICI-treated cohorts; prospective validation in trials with comprehensive biomarker collection; and the incorporation of noninvasive methylation panels as modifiers to enhance predictive power.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGFB1, MSH6
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: TGFB1/TGF-β) provides the mechanistic axis for fibrosis/immune exclusion and links directly to a histologic stromal phenotype (TSR); CORE PATH 2 (MSH6) supplies a genomic-immune counterweight (MMR deficiency → higher neoantigen load) that can recruit immune effectors despite exclusionary stroma; CORE PATH 3 (urine DNA methylation) was acknowledged as an important noninvasive readout but deliberately not placed in the core variables to keep the mechanistic hypothesis sharp.

MSH6: MSH6) supplies a genomic-immune counterweight (MMR deficiency → higher neoantigen load) that can recruit immune effectors despite exclusionary stroma; CORE PATH 3 (urine DNA methylation) was acknowledged as an important noninvasive readout but deliberately not placed in the core variables to keep the mechanistic hypothesis sharp.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	urine DNA methylation markers (e.g., SDC2, TFPI2)	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	ALP	Supporting modifier from knowledge graph
Secondary	TG	Supporting modifier from knowledge graph
Secondary	SHBG	Supporting modifier from knowledge graph
Secondary	IGF1	Supporting modifier from knowledge graph
Secondary	AI-derived biomarker signatures	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: TGF-β signaling fosters fibrotic stroma and immune exclusion in bladder cancer, with TSR serving as a practical readout of this hostile microenvironment and often predicting poor ICI response;

Prediction 2: MSH6 deficiency increases tumor mutational burden and neoantigen landscape, which can elicit robust or at least more recruitable T cell–mediated responses even within an excluded stromal context.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- urine DNA methylation markers (e.g., SDC2, TFPI2): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.